

SECTION 26 05 26

# GROUNDING AND BONDING

## TECHNICAL SPECIFICATION

|              |                                      |
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| Project      | MERIDIAN-1 Data Centre, Hall A       |
| Project code | MER-1-2026                           |
| Issue        | Issued for Construction — Revision C |
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# TABLE OF CONTENTS

|                           |                                 |
|---------------------------|---------------------------------|
| <b>PART 1 — GENERAL</b>   |                                 |
| 1.1                       | Summary                         |
| 1.2                       | References                      |
| 1.3                       | Definitions                     |
| 1.4                       | Action Submittals               |
| 1.5                       | Informational Submittals        |
| 1.6                       | Closeout Submittals             |
| 1.7                       | Quality Assurance               |
| 1.8                       | Delivery, Storage, and Handling |
| 1.9                       | Warranty                        |
| <b>PART 2 — PRODUCTS</b>  |                                 |
| 2.1                       | Manufacturers                   |
| 2.2                       | Performance Requirements        |
| 2.3                       | Equipment                       |
| 2.4                       | Components                      |
| 2.5                       | Controls and Interfaces         |
| 2.6                       | Source Quality Control          |
| 2.7                       | Factory Testing                 |
| <b>PART 3 — EXECUTION</b> |                                 |
| 3.1                       | Examination                     |
| 3.2                       | Installation                    |
| 3.3                       | Field Quality Control           |
| 3.4                       | Performance Verification        |
| 3.5                       | Commissioning                   |
| 3.6                       | Training and Closeout           |

## PART 1 — GENERAL

### 1.1 SUMMARY

- A. Provide a complete, coordinated system including engineering, manufacture, delivery, installation support, testing, commissioning, training, and closeout.
- B. Coordinate clearances, structural loads, electrical supplies, controls, fire strategy, access, and phased energisation.

### 1.2 REFERENCES

- A. Comply with the following referenced standards.
  - 1. IS 3043:2018
  - 2. IEC 60364-5-54:2011
  - 3. IEEE 80-2013

### 1.3 DEFINITIONS

- A. “Approved” means accepted for the stated stage without relieving the Contractor of responsibility.
- B. “Provide” means design, supply, install, connect, test, commission, and document.

## PART 1 — GENERAL

### 1.4 ACTION SUBMITTALS

- A. Submit project-specific product data, selected ratings, dimensions, weights, clearances, and connection requirements.
- B. Submit a clause-by-clause compliance statement identifying every exception.
- C. Submit certified calculations and performance data at scheduled conditions.

### 1.5 INFORMATIONAL SUBMITTALS

- A. Submit type-test certificates traceable to the offered construction.
- B. Submit quality certificates, calibration records, and proposed factory inspection and test plan.

### 1.6 CLOSEOUT SUBMITTALS

- A. Submit approved as-built data, final settings, test records, recommended spares, and operation and maintenance manuals.

## PART 1 — GENERAL

### 1.7 QUALITY ASSURANCE

- A. Manufacturer shall maintain an ISO 9001-certified quality system and demonstrate comparable data-centre experience.
- B. Test instruments shall have current calibration traceable to a recognised laboratory.

### 1.8 DELIVERY, STORAGE, AND HANDLING

- A. Protect equipment from moisture, shock, corrosion, and contamination.
- B. Maintain equipment-tag, purchase-order, serial-number, and source-lot traceability.

### 1.9 WARRANTY

- A. Provide the project warranty commencing at Taking Over or the date stated in the Contract, whichever governs.

## PART 2 — PRODUCTS

### 2.1 MANUFACTURERS

- A. Subject to compliance with the Contract Documents, provide the system by an approved manufacturer.
- B. Products shall be current production models with local technical support in India.
- C. Earth conductor purity shall be not less than **99.9 %**.

### 2.2 PERFORMANCE REQUIREMENTS

- A. Equipment shall operate continuously at the scheduled duty and site conditions.
- B. Selections shall include stated tolerances, fouling allowances, harmonics, diversity, and redundancy as applicable.
- C. Published catalogue values shall not replace project-condition certified data.

## PART 2 — PRODUCTS

### 2.3 EQUIPMENT

- A. Provide a complete factory-engineered assembly with coordinated ratings and interfaces.
  - 1. Buried copper grid and electrodes
  - 2. Equipment bonding conductors and test links
  - 3. Earth bars, pits and equipotential bonds
- B. Earth grid resistance shall not exceed **0.5 ohm**.

### 2.4 COMPONENTS

- A. Components shall be accessible for inspection, isolation, removal, and replacement.
- B. Safety-critical devices shall be independently identified and fail to a safe state.
- C. Internal wiring, terminals, piping, and accessories shall be permanently identified.

## PART 2 — PRODUCTS

### 2.5 CONTROLS AND INTERFACES

- A. Hardwired safety and shutdown functions shall not depend solely on the supervisory network.
- B. Provide scheduled BACnet/IP or Modbus TCP status, alarm, command, measurement, and setpoint objects.
- C. Synchronise event timestamps to the project NTP service in Asia/Kolkata.

### 2.6 SOURCE QUALITY CONTROL

- A. Perform documented incoming inspection, in-process checks, and final inspection.
- B. Record material traceability, calibrated instrument numbers, torque values, and as-left settings.



## PART 2 — PRODUCTS

### 2.7 FACTORY TESTING

- A. Submit the factory test procedure before testing.
- B. Test the offered rating and construction at the specified operating condition.
- C. Record raw measurements, calculated results, instrument details, and acceptance status.
- D. Do not claim equivalence for a different rating or condition without an approved engineering comparison.

## PART 3 — EXECUTION

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### 3.1 EXAMINATION

- A. Verify foundations, access routes, clearances, environmental conditions, and embedded services before installation.
- B. Do not proceed until unsatisfactory conditions are corrected.

### 3.2 INSTALLATION

- A. Install in accordance with approved drawings and manufacturer instructions.
- B. Install level, aligned, restrained, labelled, and accessible for operation and maintenance.
- C. Coordinate earthing, power, controls, penetrations, fire stopping, drainage, and builder's work.

## PART 3 — EXECUTION

### 3.3 FIELD QUALITY CONTROL

- A. Inspect the completed installation and record deficiencies.
- B. Perform applicable continuity, insulation, pressure, leakage, rotation, interlock, and alarm tests.
- C. Bond continuity shall not exceed **0.1 ohm**.

### 3.4 PERFORMANCE VERIFICATION

- A. Test standard shall be **IS 3043:2018**.
- B. Record measured values, instrument uncertainty, acceptance limits, and status.

## PART 3 — EXECUTION

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### 3.5 COMMISSIONING

- A. Execute L2 site acceptance, L3 pre-functional, L4 functional performance, and L5 integrated systems testing.
- B. Coordinate test prerequisites, witnesses, temporary loads, trend logs, and restoration.

### 3.6 TRAINING AND CLOSEOUT

- A. Train operations personnel using the installed equipment.
- B. Submit signed test records, final settings, training attendance, and approved closeout documents.

**END OF SECTION**